

Assessment Brief - Coursework

Academic Year	2025-26
Semester	1
Module Number	CM 4607
Module Title	Computational Intelligence
Assessment Method	Coursework
Deadline (time and date)	Part A -29 th December 2025 Part B – 9 th December 2025 (tentative)
Submission	Part A- Assessment Dropbox in the Module Study Area in CampusMoodle. A submission of code and report followed by demonstration and viva (failing to attend viva will set the grade to fail) Part B – 90 minutes in-class test
Word Limit	4000
Use of Generative Artificial Intelligence (AI) text	IS authorised.
Module Co-ordinator	Fathima Farhath

What knowledge and/or skills will I develop by undertaking the assessment?

1. Realize real-world industry application in terms of Evolutionary computing concepts with the support of Machine Learning and other associated algorithms in Artificial Intelligence.
2. Develop holistic view of very complex problems and thereby modularize the problems into solvable modules that can be solved by the Machine learning and EC algorithms together.
3. Design and implement the realized solution using given libraries and discuss the effectiveness of the developed solution.

On successful completion of the assessment students will be able to achieve the following Learning Outcomes:

Part A:

What knowledge and/or skills will I develop by undertaking the assessment?

1. LO2 Develop an adaptive intelligent system for a given application.
2. LO4 Execute selected computational intelligence algorithms in adaptive intelligent systems

Part B

3. LO1 Examine problems in optimisation and machine learning to select suitable solution techniques.
4. LO3 Illustrate current application areas of adaptive intelligent systems.

Please also refer to the Module Descriptor, available from the module Moodle study area.

What is expected of me in this assessment?

Task(s) - content

Part A:

Task 1: Identify an industry application in an appropriate scale level where you can demonstrate the applicability of the evolutionary algorithms.

Task 2: Discuss and justify the necessity of solving the identified problem using evolutionary algorithm (Indicate why it is not possible to solve as a typical IT solution or numerical optimization)

Task 3: Draw the holistic view of the artificial intelligence solution you provide and thereby develop the modularized diagram to realize the holistic view.

Task 4: Define the methodology by defining the chromosome structure, fitness function, constraints.

Task 5: Develop the methodology using appropriate API framework and demonstrate the findings.

Part B:

Participate in the in-class test and demonstrate the knowledge and understanding on the theoretical aspects of computational intelligence

Task(s) - format

Submission: Software and Brief Report consist of Problem Definition, Approach (Designing of EC algorithm) Maximum 4000 words.

How will I be graded?

A number of subgrades will be provided for each criterion on the feedback grid which is specific to the assessment.

The overall grade for the assessment will be calculated using the algorithm below*.

A	At least 50% of the subgrades to be at Grade A, at least 75% of the subgrades to be at Grade B or better, and normally 100% of the subgrades to be at Grade C or better.
B	At least 50% of the subgrades to be at Grade B or better, at least 75% of the subgrades to be at Grade C or better, and normally 100% of the subgrades to be at Grade D or better.
C	At least 50% of the subgrades to be at Grade C or better, and at least 75% of the subgrades to be at Grade D or better.
D	At least 50% of the subgrades to be at Grade D or better, and at least 75% of the subgrades to be at Grade E or better.
E	At least 50% of the subgrades to be at Grade E or better.
F	Failing to achieve at least 50% of the subgrades to be at Grade E or better.
NS	Non-submission.

*If the word count is above the specified word limit by more than 10% or the submission contains an excessive use of text within tables, the grade for the submission will be reduced to the next lowest grade.

Feedback grid

GRADE	A	B	C	D	E	F
DEFINITION / CRITERIA (WEIGHTING)	EXCELLENT Outstanding Performance	COMMENDABLE/VERY GOOD Meritorious Performance	GOOD Highly Competent Performance	SATISFACTORY Competent Performance	BORDERLINE FAIL	UNSATISFACTORY Fail
PART -A						
Problem Identification (1 subgrades)	Appropriate introduction and briefing on an industry application where problem has become very complex in terms of the number of possible permutations to be considered for optimization are very high and to solve the entire problem the support of machine learning techniques is required.	Appropriate introduction and briefing on an industry application where problem has become very complex in terms of the number of possible permutations to be considered for optimization are very large and need the support of machine learning techniques to solve the problem.	Appropriate introduction and briefing on an industry application where problem has become complex in terms of the number of possible permutations to be considered for optimization are moderately large and need the support of machine learning techniques to solve the problem.	Appropriate introduction and briefing on an industry application where problem has become complex in terms of the number of possible permutations to be considered for optimization are moderately large.	Appropriate introduction and briefing on an industry application where problem is not complex in terms of number of permutations to be considered for optimization.	Introduction to a toy problem rather than an industry problem
Justification for use of GA (1 subgrades)	Appropriate justification of the complexity of the problem in two aspects: the need of simulating the future under risk and uncertainty and optimize the objective function on the simulated scenario.	Appropriate justification on the complexity of the problem by discussing how the constraints and conditions applied on the objective function.	Appropriate discussion highlighting the need of implementing evolutionary computing algorithm on the simulated environment (using machine learning) on a such moderately complex environment.	Appropriately justify the need of evolutionary algorithm on the moderately complex environment and justify why it is not solvable using numerical optimization technique	Discuss the selection of an industry application to solve using evolutionary computing rather than numerical optimization	No significant justification was made to highlight the need of using evolutionary algorithms.

GRADE	A	B	C	D	E	F
DEFINITION / CRITERIA (WEIGHTING)	EXCELLENT Outstanding Performance	COMMENDABLE/VERY GOOD Meritorious Performance	GOOD Highly Competent Performance	SATISFACTORY Competent Performance	BORDERLINE FAIL	UNSATISFACTORY Fail
Modularized diagram (1 subgrades)	Provide detail drawing of the modularized diagram by highlighting the salient input features and output of each module. And thereby integrating all the module to demonstrate the holistic approach of the solution.	Provide a drawing of the modularized diagram by showing the input features and output of each module. And thereby integrating all the module to demonstrate the holistic approach of the solution.	Provide detail drawing of the modularized diagram by highlighting the salient input features and output of each module.	Some sketches on holistic view of the solution and modularized diagram with some input features and outputs are identified.	A modularized diagram with some input features and outputs are identified.	No comprehensive diagrams have produced to understand the overall approach of the solution.
Implementation (3 subgrades)	An end-to-end prototype which simulates the methodology presented.	A prototype which simulates the core-components of the methodology presented.	A somewhat integrated software that simulates the core-component of the methodology presented.	A set of components of software modules which executes the given task.	Only few software components that works alone	No proper modules of software can be found.
PART - B						
In-Class Test (4 subgrades)	Mark >80%	65% <= mark < 80%	45% <= mark < 65%	35% <= mark < 45%	35% <= mark < 25%	Mark <25

Coursework received late, without valid reason, will be regarded as a non-submission (NS) and one of your assessment opportunities will be lost.

For Part – A, the assessment of each subcomponent will be based not only on the quality and clarity of the written report, but also on the student's ability to effectively justify, defend, and communicate their work during the viva. Demonstrated understanding, critical reasoning, and the capacity to respond to questions will be the key criteria.

What else is important to my assessment?

What is the Assessment Word Limit Statement?

It is important that you adhere to the Word Limit specified above. The Assessment Word Limit Statement can be found in Appendix 2 of the [RGU Assessment Policy](#). It provides detail on the purpose, setting and implementation of wordage limits; lists what is included and excluded from the word count; and the penalty for exceeding the word count.

What's included in the word count?

The table below lists the constituent parts which are included and excluded from the word limit of a Coursework; more detail can be found in the full Assessment Word Limit Statement. Images will not be allowed as a mechanism to circumvent the word count.

Excluded	Included
Cover or Title Page	Main Text e.g. Introduction, Literature Review, Methodology, Results, Discussion, Analysis, Conclusions, and Recommendations
Executive Summary (Reports) or Abstract	Headings and subheadings
Contents Page	In-text citations
List of Abbreviations and/or List of Acronyms	Footnotes (relating to in-text footnote numbers)
List of Tables and/or List of Figures	Quotes and quotations written within “...”
Tables – mainly numeric content	Tables – mainly text content
Figures	
Reference List and/or Bibliography	
Appendices	
Glossary	

What are the penalties?

The grade for the submission will be reduced to the next lowest grade if:

- The word count of submitted work is above the specified word limit by more than 10%.
- The submission contains an excessive use of text within Tables or Footnotes.

What else is important to my assessment?

What is plagiarism?

Plagiarism is “the practice of presenting the thoughts, writings or other output of another or others as original, without acknowledgement of their source(s) at the point of their use in the student’s work. All materials including text, data, diagrams or other illustrations used to support a piece of work, whether from a printed publication or from electronic media, should be appropriately identified and referenced and should not normally be copied directly unless as an acknowledged quotation. Text, opinions or ideas translated into the words of the individual student should in all cases acknowledge the original source” ([RGU 2022](#)).

What is collusion?

“Collusion is defined as two or more people working together with the intention of deceiving another. Within the academic environment this can occur when students work with others on an assignment, or part of an assignment, that is intended to be completed separately” ([RGU 2022](#)).

For further information please see [Academic Integrity](#).

What if I'm unable to submit?

- The University operates a [Fit to Sit Policy](#) which means that if you undertake an assessment then you are declaring yourself well enough to do so.
- If you require an extension, you should complete and submit a [Coursework Extension Form](#). This form is available on the RGU [Student and Applicant Forms](#) page.
- Further support is available from your Course Leader.

What additional support is available?

- [RGU Study Skills](#) provide advice and guidance on academic writing, study skills, maths and statistics and basic IT.
- [RGU Library guidance on referencing and citing](#).
- [The Inclusion Centre: Disability & Dyslexia](#).
- Your Module Coordinator, Course Leader and designated Personal Tutor can also provide support.

What are the University rules on assessment?

The University Regulation '[A4: Assessment and Recommendations of Assessment Boards](#)' sets out important information about assessment and how it is conducted across the University.